

# Eric Julianto

AI Systems Engineer | Applied AI, GenAI, RAG, Computer Vision, and Cloud-Native Delivery

Jakarta, Indonesia | [ericjulianto00@gmail.com](mailto:ericjulianto00@gmail.com) | +62 812 9985 1190

[algonacci.github.io](https://algonacci.github.io) | [linkedin.com/in/eric-julianto](https://linkedin.com/in/eric-julianto) | [Google Scholar](https://scholar.google.com/citations?user=...) | [github.com/algonacci](https://github.com/algonacci)

## Professional Summary

---

AI Systems Engineer building at the intersection of machine learning, generative AI, and production-grade infrastructure. Background spans LLM applications, RAG pipeline design, vision language models, forecasting research, and cloud-native deployment, with infrastructure knowledge across Dockerized services, observability stacks (Prometheus, Grafana, Loki, Tempo), messaging layers, and distributed systems. Positioned for roles requiring both model understanding and systems execution — AI Engineer, GenAI Engineer, ML Engineer, and AI Systems Engineer tracks.

## Core Skills

---

**Programming Languages:** Python, JavaScript, TypeScript, Rust, Go, PHP, Elixir, Dart

**AI/ML:** Machine Learning, Deep Learning, Computer Vision, NLP, Time Series Forecasting, Applied Research, Model Evaluation

**Generative AI:** LLM, VLM, Multi-modal, RAG, Multi-Agent Systems, MCP, A2A, Prompt Engineering, LangChain, Ollama

**Frameworks:** FastAPI, Flask, PyTorch, React, Laravel, Actix, Phoenix

**Cloud & DevOps:** Docker, Kubernetes, AWS, GCP, Cloudflare, PM2, GitHub Actions, GitLab CI

**Observability:** Prometheus, Grafana, Loki, Tempo, Telemetry, Log Aggregation, Monitoring

**Databases:** PostgreSQL, MySQL, MariaDB, Redis, MinIO

**Infrastructure & Systems:** Linux, Ubuntu, Red Hat, RabbitMQ, VPN, Virtualized Environments

## Experience

---

### Bangkit Academy

Feb 2023 – Jan 2025

*Machine Learning Mentor & Advisor*

- Guided machine learning learners through model-building workflows, assignments, and applied project execution in a structured cohort environment.
- Tracked learner progress, identified technical blockers early, and delivered targeted support across core ML concepts and implementation details.
- Collaborated with instructors to refine learning materials and improve assignment clarity for a more execution-focused learning path.

### Dibilabs by Dibimbing.id

Mar 2023 – Jun 2023

*SEO Intern*

- Performed keyword and competitor analysis to identify content opportunities and improve search visibility across target topics.
- Optimized on-page content structure, metadata, and media annotations using analytics-backed SEO workflows.
- Monitored traffic, rankings, and keyword performance using web analytics and SEO tooling to support iterative content improvements.

*Mentor*

- Delivered mentorship across data science, machine learning, deep learning, and computer vision topics in a structured community setting.
- Tracked learner progress and reported development updates to program organizers.
- Acted as a technical mentor and learning guide for students progressing through applied AI material.

**ZettaByte Pte Ltd.**

May 2022 – Aug 2022

*AI Developer*

- Built AI application components including a student focus-level measurement system, question generation workflow, and question answering system.
- Deployed chatbot functionality with database integration as part of a broader AI product implementation.
- Monitored development execution, tracked progress, and followed through on delivery plans during implementation cycles.

**Publications & Research**

---

- **Improving Support Vector Machine Performance with Binary Gaussian Improved Whale Optimization Algorithm: A Case Study on Diabetes Data** — *BAREKENG: Jurnal Ilmu Matematika dan Terapan* (SINTA 2, Scopus Q4).
- **Optimizing Heart Attack Diagnosis Using Random Forest with Bat Algorithm and Greedy Crossover Technique** — *BAREKENG: Jurnal Ilmu Matematika dan Terapan* (SINTA 2, Scopus Q4).
- **Adaptive Multi-Criteria Re-Ranking for Hybrid Semantic Information Retrieval in Metadata-Rich Collections** — Scopus-indexed IEEE conference publication.
- **Automatic Plate Number Recognition (APNR) Using Deep Learning with YOLOv8 and TrOCR** — Scopus-indexed IEEE conference publication.
- **Robust Dominant Color Extraction Using Saliency-Weighted K-Means with Power-Law Modulation for Complex Backgrounds** — Scopus-indexed IEEE conference publication.

**Selected Projects**

---

**Omniflow — AI-Native Modular Enterprise Platform**

Flagship project

- Designed an AI-native modular enterprise platform combining intelligent workflow orchestration, LLM-powered automation, RAG pipelines, and MCP-based agentic orchestration for enterprise process automation.
- Built distributed infrastructure spanning Dockerized services, PostgreSQL, Redis, RabbitMQ, MinIO, and an observability stack (Prometheus, Grafana, Loki, Tempo) with telemetry-driven monitoring and cloud-native deployment on Cloudflare.
- Integrated systems-thinking across modular ERP modules, AI-native automation pipelines, observability tooling, and production-oriented platform architecture with end-to-end service interoperability.
- Explored agentic system experimentation and MCP-based orchestration patterns for enterprise workflow automation, enabling multi-step task coordination and intelligent decision routing.

Tech stack: Python, JavaScript, Docker, Linux, PostgreSQL, Redis, RabbitMQ, MinIO, Grafana, Loki, Tempo, LangChain, RAG, MCP, Cloudflare, PM2

## KTP OCR with VLM

Selected project

- Built an OCR pipeline for extracting Indonesian ID card (KTP) data using a Vision Language Model, producing structured JSON output for e-KYC workflows.
- Engineered handling for noisy text, multi-field column layouts, and varied image quality to support downstream identity verification pipelines.
- Designed the extraction pipeline with inference-aware processing and structured output validation for production-oriented document intelligence.

Tech stack: Vision Language Model, Python, Structured Extraction, Document Intelligence, e-KYC

## Finbot

Selected project

- Developed a financial analysis web application for ^JKSE and S&P 500 with real-time data retrieval, time series forecasting, and API-driven market data integration.
- Integrated news aggregation, YouTube insight extraction, and LLM-powered interactive Q&A for comprehensive financial analysis and decision support.
- Combined forecasting models with real-time data pipelines to deliver actionable market intelligence through a unified interface.

Tech stack: Python, Time Series Forecasting, Financial APIs, LLM, RAG, Web Application

## Document Translator

Selected project

- Built a document translation pipeline with LLM-powered translation, supporting multi-format output downloads as PDF or TXT packaged in ZIP.
- Implemented document-grounded chat functionality using retrieval-based context for follow-up questions on translated content.
- Designed the workflow around document ingestion, chunking, translation orchestration, and output packaging for scalable document processing.

Tech stack: LLM, RAG, Document Processing, PDF Generation, Translation Pipeline

## Marketplace Product Recommendation System

Selected project

- Built a multi-modal recommender system for marketplace products combining keyword-based text retrieval and image similarity search.
- Integrated visual and textual feature extraction to improve product discovery, relevance ranking, and cross-modal recommendation accuracy.

Tech stack: Python, Machine Learning, Computer Vision, Recommendation Systems, Vector Retrieval

## Palm Oil Ripeness Detection

Selected project

- Developed a specialized web application for assessing palm oil fruit ripeness with real-time image analysis and ML-based classification.
- Delivered harvesting optimization insights and interactive tools for plantation yield tracking and operational efficiency improvement.

Tech stack: Machine Learning, Computer Vision, Real-Time Inference, Web Application

## AI Interview

Selected project

- Built an automated interview platform using LLM-driven language and context analysis to evaluate candidate responses.
- Designed the evaluation pipeline with structured prompt orchestration and context-aware scoring for consistent, AI-driven candidate assessment.

Tech stack: LLM, NLP, Prompt Orchestration, Web Application, Recruitment Tech

## Education

---

### Universitas Esa Unggul

Mar 2025 – Present

#### *Computer Science*

- Area of interest: Artificial Intelligence, Machine Learning, Computational Finance
- GPA: 4.00

### Universitas Bunda Mulia

Aug 2019 – May 2024

#### *Hospitality and Tourism*

- Area of interest: Tourism Management & Development and Risk Management
- GPA: 3.67
- TOEFL equivalent score: 567
- Recognition: Mahasiswa Berprestasi

### Bangkit Academy led by Google, Tokopedia, Gojek & Traveloka

Jan 2022 – Jul 2022

#### *Machine Learning*

- Area of interest: Machine Learning, Artificial Intelligence, Deployment
- GPA equivalent score: 4.0

### AI for Startup | Orbit Future Academy

Aug 2021 – Jan 2022

#### *Data Science*

- Area of interest: Data Science, Machine Learning, Artificial Intelligence
- GPA equivalent score: 4.0
- Top 16 Finalists Habibie Techfest

## Speaking

---

- “Shifting Career To Data Science” — [Webinar Coding.id](#).
- “Kickstart Career in Data Science” — [HIMTI UIN Jakarta](#).
- “NVIDIA Jetson Nano for AI” — [Indonesia AI Chit-chat](#).
- “IoT Sederhana Menggunakan Arduino” — [Webinar Series Lomba THINC AI BISA AI](#).
- “UI/UX Design with Figma and Other Tools” — [Webinar Series Lomba THINC AI BISA AI](#).
- “Kenalan Yuk Sama Intel Galileo” — [Webinar Series Lomba THINC AI BISA AI](#).
- “Artificial Intelligence 101” — [Forum Ilmiah Politeknik Astra](#).
- “Generative AI Algorithms in Mobile Application” — Tech Talk at GDSC UMS.
- “AI Integration in IoT World” — AMIKOM Computer Club.
- “Deep Dive Into the Deep Learning” — [Tech Talk at GDSC UTM](#).
- “AI sebagai alat bantu untuk meningkatkan kreativitas manusia” — [Tech Talk at Universitas Andalas](#).
- “Pemanfaatan Artificial Intelligence (AI) Dalam Penulisan Karya Ilmiah” — [National Seminar at UM-SIDA](#).
- “Text2Text Generation with Gemini” — [AI Workshop at UPN Jatim](#).
- “GDSC Forward STT NF” — [Tech Talk at STT NF](#).
- “Designing Intelligent Recommendation Systems with Google Vertex AI” — [Build with AI: Advance Workshop at GDGoC STT Terpadu Nurul Fikri](#).
- “AI and Environment: Paths to Regenerative Sustainability” — [Tech Talk at UPB Cikarang](#).
- “From Code to Production: A Hands-On MLOps Workshop” — BCC FILKOM UB.